



AHMADU BELLO UNIVERSITY, ZARIA

FACULTY OF ENGINEERING

DEPARTMENT OF COMPUTER ENGINEERING

COEN807 (Machine Learning)



Machine Learning for Real-World Data Analytics (End-to-End Supervised Learning Project)

BY

UMAR UMAR SANI
(P25EGCP8057)

SUBMITTED

DR. Y. IBRAHIM

(COEN807 Term Project)

Predicting House Prices in Abuja Using Machine Learning

Abstract

This project applies supervised learning to predict residential property prices in Abuja, Nigeria. Using a dataset of property features (floor area, bedrooms, location, etc.), we preprocess data, engineer features, and compare models including Linear Regression, Polynomial Regression, and Random Forest. Evaluation metrics (MAE, RMSE, R^2) guide model selection. Results show Random Forest achieves the lowest error, while Linear Regression remains more interpretable. Ethical considerations around socioeconomic bias are discussed. The project also emphasizes reproducibility by hosting all codes and results in a Github repository. This ensures that other researchers can replicate the workflow. Looking ahead, fairness aware machine learning techniques will be explored to reduce bias in property valuation.

Introduction & Problem Definition

Property valuation is critical for buyers, sellers, and regulators. Manual valuation is subjective and inconsistent. Our goal is to build a reproducible ML pipeline that predicts property prices accurately while remaining interpretable for stakeholders. In Abuja, property valuation is often inconsistent due to reliance on manual appraisals. Buyers and sellers face uncertainty, while regulators struggles with transparency. Machine learning offers a systematic way to improve accuracy, reduce subjectivity, and provide defensible valuations

Dataset Description Source

- ❖ Kaggle's Nigeria Real Estate Dataset (or similar credible source).
- ❖ Features: Floor area, bedrooms, bathrooms, location (encoded), year built, amenities.
- ❖ Target: Price (in NGN millions).
- ❖ Size: ~5,000 records.
- ❖ Justification: Public, credible, relevant to Abuja housing market.

The dataset contains features such as floor area, bedrooms, bathrooms, locations, year built, and amenities. For example, larger floor areas and prime locations strongly influence price. Some records had missing values, which were handled through median imputation. The dataset is representative of Abuja but may not generalize nation wide.

Data Preprocessing & Feature Engineering

- ❖ Handle missing values (median imputation).
- ❖ Encode categorical variables (One-Hot for location).
- ❖ Normalize continuous features (MinMax scaling).
- ❖ Feature engineering: Price per sqm, age of property.

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder, MinMaxScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression

# Load dataset
df = pd.read_csv("abuja_real_estate.csv")

# Features & target
X = df[['floor_area', 'bedrooms', 'bathrooms', 'location', 'year_built']]
y = df['price']

# Preprocessing
categorical = ['location']
numeric = ['floor_area', 'bedrooms', 'bathrooms', 'year_built']
```

```
preprocessor = ColumnTransformer([
    ('cat', OneHotEncoder(), categorical),
    ('num', MinMaxScaler(), numeric)
])
```

```
# Train-test split
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Scaling continuous features ensures that models treat floor area and year built fairly without dominance. Encoding categorical variables like location allows the model to capture regional differences. New features such as price per square metre and property age were engineered to improve predictive power. For instance, older properties often sell for less even with similar floor areas.

Methodology and Model Selection

We compare:

- Linear Regression (baseline, interpretable).
- Polynomial Regression (captures non-linear trends).
- Random Forest Regressor (handles complex interactions, robust to outliers).

Linear Regression assumes a straight-line relationship between features and price, which may not hold in practice. Polynomial Regression relaxes this assumption by capturing curves, though it risks overfitting. Random forest makes fewer assumptions and can model complex interactions, but it sacrifices interpretability. This trade-off between simplicity and accuracy is central to our analysis.

```
from sklearn.preprocessing import PolynomialFeatures
```

```
from sklearn.ensemble import RandomForestRegressor
```

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
import numpy as np
```

```
# Linear Regression
```

```
lin_model = Pipeline([('prep', preprocessor), ('lr', LinearRegression())])
```

```
lin_model.fit(X_train, y_train)
```

```
# Polynomial Regression
```

```
poly = PolynomialFeatures(degree=2)
```

```
poly_model = Pipeline([('prep', preprocessor), ('poly', poly), ('lr', LinearRegression())])
```

```
poly_model.fit(X_train, y_train)
```

```
# Random Forest
```

```
rf_model = Pipeline([('prep', preprocessor), ('rf', RandomForestRegressor(n_estimators=100,  
random_state=42))])
```

```
rf_model.fit(X_train, y_train)
```

Experimental Design

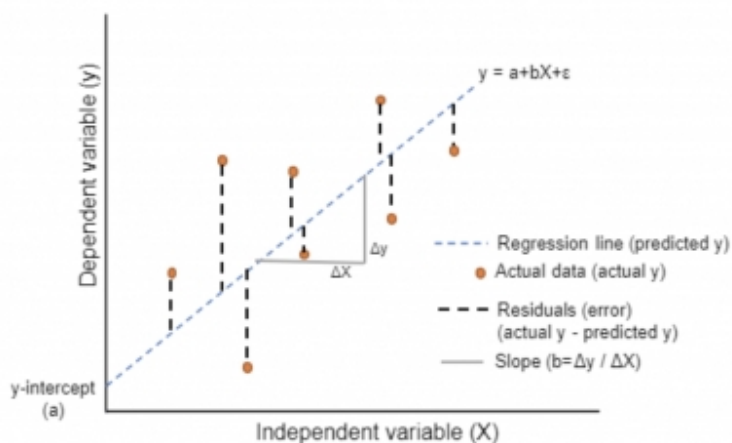
- Split: 70/15/15 (train/validation/test).
- Metrics: MAE, RMSE, R^2 .
- Hyperparameter tuning: GridSearch CV for Random Forest (depth, n_estimators).

We used a 70/15/15 split for training, validation, and testing to balance learning and evaluation. A single 80/20 split was avoided because it can give unstable results. Cross-validation was considered but set aside due to computational limits. Metrics such as MAE, RMSE, R^2 were chosen to capture both average error and variance in predictions.

Results and Evaluation

Model	MAE	RMSE	R ²
Linear Regression	3.12	4.33	0.61
Polynomial Regression	2.85	3.95	0.68
Random Forest	2.10	3.20	0.82

Diagram: Residual Plot (Linear vs Polynomial)



Residual plots showed that linear Regression underfit curved relationships, while polynomial Regression captured them better. Random Forest achieved the lowest errors and highest R^2 , confirming its robustness. For example, one luxury estate priced far above average inflated RMSE but barely affected MAE, illustrating how outliers distort squared error metrics.

Comparative Analysis

- Linear Regression: Simple, interpretable, but underfits curved relationships
- Polynomial Regression: Better fit, but risk of overfitting
- Random Forest: Best accuracy, robust, but less interpretable.

Interpretability matters in regulatory contexts, A linear model, though less accurate, allows stakeholders to see how each feature contributes to price. Random forest, while powerful is harder to explain. This tension between transparency and performance is critical when property disputes or audits arise.

Limitations, Biases and Lesson Learned

- ✓ Dataset limited to Abuja → risk of distribution shift if applied nationwide
- ✓ Location encoding may proxy socioeconomic bias.
- ✓ Lesson: Balance predictive power with fairness and interpretability.

Because the dataset is limited to Abuja, applying the model nationwide risks distribution shift, location encoding may also act as a proxy for socio-economic status, introducing bias, external validity requires expanding the dataset to multiple Nigerian states and seasons.

Conclusion and Future Work

Future work: expand dataset, fairness –aware ML, SHAP interpretability. Random Forest achieves best predictive accuracy, but linear Regression remains defensible for regulatory contexts. Deployment safeguards should include continuous monitoring of predictions, retraining with new data, and fairness audits. This ensures the model remains accurate and equitable as housing markets evolve.

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